

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

5

Code of Conduct:

I v.surya satvika(192524254) __ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:v.s.satvika

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

1(b) Handling Class Imbalance

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

1(a) Inductive Learning Approach

Inductive learning learns general patterns from a set of labelled examples and uses these learned patterns to predict the class of new, unseen patients.

For this healthcare dataset, the problem can be formulated as a binary classification problem. The model predicts whether a patient is diabetic or non-diabetic.

Target Variable:

- Diabetes Status
- 0 – Non-diabetic
- 1 – Diabetic

Important Features:

1. **Glucose Level:** High glucose levels are strongly associated with diabetes and therefore make an important predictor.
2. **BMI:** A higher BMI can be associated with increased diabetes risk.
3. **Age:** Diabetes risk generally changes with age.
4. **Blood Pressure:** Abnormal blood pressure provides useful information about the patient's metabolic health.
5. **Cholesterol:** Cholesterol levels provide additional information about the patient's health condition.
6. **Family History:** Family history can indicate increased genetic susceptibility to diabetes.

Inductive Learning Process:

Patient Data → Preprocessing → Feature Selection → Training Model → Learn Patterns →
New Patient → Diabetes Prediction

The model learns relationships from historical patient records and generalises the learned patterns to make predictions for unseen patients.

1(b) Handling Class Imbalance

The dataset contains more non-diabetic patients than diabetic patients, creating class imbalance. If this imbalance is ignored, the model may favour the majority class and incorrectly classify diabetic patients as non-diabetic.

Suitable Strategy: SMOTE

I would use SMOTE (Synthetic Minority Over-sampling Technique). SMOTE generates synthetic examples of the minority class instead of simply duplicating existing diabetic records.

Procedure:

1. Separate diabetic and non-diabetic observations.
2. Identify diabetic cases as the minority class.
3. Apply SMOTE only to the training data.
4. Generate synthetic diabetic samples.
5. Train the classifier using the balanced training set.
6. Keep the test set unchanged for unbiased evaluation.

Why SMOTE?

- It increases the representation of diabetic cases.
- It reduces majority-class bias.
- It can improve recall for diabetic patients.
- It is preferable to simply duplicating the same minority records.
- In medical diagnosis, missing a diabetic patient is more serious than producing an additional false alarm.

Therefore, SMOTE is a suitable strategy for this medical classification problem.

1(c) 70:30 Data Split and Preprocessing

The dataset is divided into:

- 70% – Training data
- 30% – Testing data

The 70:30 split provides enough data for the model to learn while retaining a sufficiently large independent test set for evaluation.

For a medical application, the split should preferably be stratified, so that the proportion of diabetic and non-diabetic patients remains similar in both sets.

Preprocessing Steps:

1. **Missing Value Handling:** Missing numerical values can be replaced using suitable statistical methods such as median imputation.

2. **Encoding:** Categorical features such as Family History can be converted into numerical form:
 - No $\rightarrow$ 0
 - Yes $\rightarrow$ 1
3. **Normalisation:** Numerical features such as Age, Blood Pressure, Cholesterol, Glucose and BMI can be standardised using:

$$z = (x - \mu) / \sigma$$

This gives the features a comparable scale.

4. **Class Balancing:** SMOTE should be applied after splitting and only to the training set to avoid data leakage.

Normalisation is especially important for statistical models such as Logistic Regression, while Decision Trees are generally less sensitive to feature scaling.

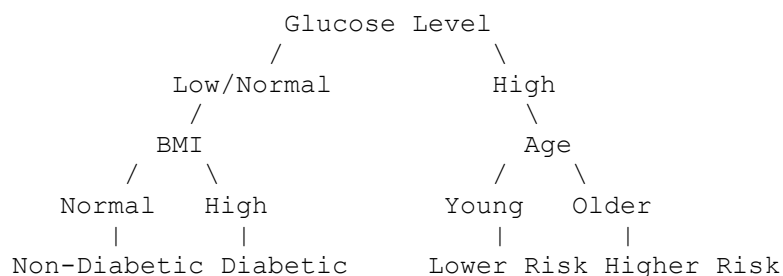
Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

2(a) Decision Tree Classifier

A Decision Tree recursively divides the dataset using feature conditions until it reaches a prediction.

First Three Levels:



Root Node:

Glucose Level is a strong candidate for the root node because it is highly informative for separating diabetic and non-diabetic patients.

The actual root should be selected by calculating the highest Information Gain or lowest Gini impurity from the training dataset.

Information Gain:

$$IG(S,A) = Entropy(S) - \sum_v (|S_v|/|S|) Entropy(S_v)$$

The feature producing the highest Information Gain is selected.

Gini Index:

$$Gini = 1 - \sum p_i^2$$

The feature producing the lowest Gini impurity is preferred.

Therefore, Glucose Level can be selected as the root if it produces the highest Information Gain or lowest Gini value in the actual dataset.

2(b) Model Evaluation

The Decision Tree should be evaluated using the following measures:

1. Accuracy

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

2. Precision

$$Precision = TP / (TP + FP)$$

3. Recall

$$Recall = TP / (TP + FN)$$

4. F1-Score

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$$

False Negative:

A False Negative (FN) occurs when:

- Actual patient = Diabetic
- Model prediction = Non-diabetic

This is particularly important because a missed diabetes case could delay diagnosis and treatment.

Methods to Reduce False Negatives:

1. Use SMOTE.
2. Apply class weights.
3. Tune the classification threshold.
4. Perform hyperparameter tuning.
5. Use cost-sensitive learning.
6. Monitor the confusion matrix.

Clinically, reducing false negatives is important because the objective is early diagnosis and missed diabetic cases may not receive timely medical attention.

2(c) Pre-Pruning vs Post-Pruning

Pre-Pruning:

Pre-pruning stops the tree from becoming unnecessarily complex during training.

Examples:

- Maximum tree depth
- Minimum samples per split
- Minimum samples per leaf

Advantages:

- Faster training
- Smaller tree
- Reduces overfitting

Disadvantage:

The tree may stop growing too early and underfit.

Post-Pruning:

Post-pruning first builds a larger tree and then removes branches that do not sufficiently improve performance.

The assessment specifies cost-complexity pruning.

$$R_{\alpha}(T) = R(T) + \alpha|T|$$

Where:

- $R(T)$ = Error of the tree
- $|T|$ = Number of terminal nodes

- α = Complexity parameter

Comparison:

Pre-pruned Tree – Simple and fast, but may underfit.

Post-pruned Tree – Removes unnecessary branches but requires additional evaluation.

Unpruned Tree – Can capture complex patterns but is more likely to overfit.

Clinical Deployment:

A pruned Decision Tree is generally more appropriate for deployment because it is simpler, easier to interpret and less likely to overfit. However, the final model should be selected based on validation/test performance, particularly Recall, F1-score and clinical safety rather than Accuracy alone.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

3(a) Statistical Learning for Risk Stratification

A suitable statistical learning model for this dataset is Logistic Regression. It can be used to predict the probability that a patient is diabetic or non-diabetic.

Logistic Regression is suitable because the target variable, Diabetes Status, is binary.

The model uses features such as Age, Blood Pressure, Cholesterol, Glucose Level, BMI and Family History.

The performance of Logistic Regression can be compared with the Decision Tree using Accuracy and AUC-ROC.

Accuracy measures the proportion of correctly classified patients.

AUC-ROC measures how well the model distinguishes between diabetic and non-diabetic patients. A higher AUC indicates better discrimination.

Comparison:

Decision Tree – Accuracy and AUC-ROC are calculated using the test data.

Logistic Regression – Accuracy and AUC-ROC are calculated using the test data.

A statistical model such as Logistic Regression is preferable when the dataset has a relatively simple relationship between features and the outcome, when probability estimates are important, and when interpretability is required.

Logistic Regression is also useful when the relationship between predictors and the outcome can be represented effectively using a statistical model.

3(b) Feature Importance Analysis

Feature importance analysis is used to identify which input variables contribute most to predicting diabetes.

For the Decision Tree, feature importance can be obtained from the reduction in impurity produced by each feature. Features that provide larger reductions in impurity receive higher importance.

For Logistic Regression, the coefficients of the model can be examined. Features with larger absolute coefficient values have a greater influence on the prediction, assuming the numerical features have been appropriately scaled.

Top Three Predictors:

1. Glucose Level
2. BMI
3. Age

These features are expected to have strong predictive value because they provide useful information about diabetes risk.

Agreement Between Models:

The two models may agree on some important predictors but do not necessarily have to produce exactly the same ranking. This is because Decision Trees measure feature importance based on impurity reduction, whereas Logistic Regression measures the influence of features through model coefficients.

If both models identify Glucose Level, BMI and Age among the top predictors, it provides stronger evidence that these variables are useful for diabetes risk prediction.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

4(a) MDP for Treatment Recommendation

The treatment recommendation problem can be modelled as a Markov Decision Process (MDP) because the AI agent observes the patient's current health condition, selects an action and receives a reward based on the resulting health improvement.

1. States

States represent different stages of the patient's health condition.

- S1 – Low Risk: Patient has relatively healthy measurements.
- S2 – Moderate Risk: Patient has some abnormal health indicators.
- S3 – High Risk: Patient has multiple abnormal indicators.
- S4 – Controlled: Patient's health indicators have improved.

States are important because the agent needs to understand the current health condition before selecting an action.

2. Actions

The available actions are:

- Diet
- Exercise
- Medication
- Monitor

The agent selects one action depending on the patient's current state.

3. Reward Function

The reward represents the improvement in the patient's health condition.

- Positive reward – Health condition improves.
- Zero reward – Health condition remains approximately unchanged.
- Negative reward – Health condition becomes worse.

The reward encourages the agent to learn actions that lead to better health outcomes.

4. Transition Dynamics

Transition dynamics describe how the patient's health state changes after an action.

For example:

High Risk + Exercise → Moderate Risk

Moderate Risk + Diet → Controlled

The transition depends on the patient's current state and the selected action.

Therefore, the MDP can be represented as:

State → Action → Reward → New State

This allows the reinforcement learning agent to learn suitable treatment recommendations over multiple patient-state transitions.

4(b) Q-Learning for Treatment Recommendation

Q-Learning is a model-free reinforcement learning algorithm that learns the best action to take in each state by updating a Q-value table.

The Q-value represents the expected future reward for taking a particular action in a particular state.

Q-Learning Update Equation:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

- $Q(s,a)$ = Current Q-value
- α = Learning rate
- r = Reward received
- γ = Discount factor
- s' = Next state
- $\max Q(s',a')$ = Maximum future Q-value

Example Q-Table:

State	Diet	Exercise	Medication	Monitor
Low Risk	0	0	0	0

State	Diet	Exercise	Medication	Monitor
Moderate Risk	0	0	0	0
High Risk	0	0	0	0

Five Patient Episodes:

1. Episode 1: High Risk → Exercise → Moderate Risk
2. Episode 2: Moderate Risk → Diet → Controlled
3. Episode 3: High Risk → Medication → Moderate Risk
4. Episode 4: Moderate Risk → Exercise → Controlled
5. Episode 5: High Risk → Monitor → High Risk

During each episode, the Q-table is updated according to the received reward and the next state.

Convergence Criterion:

The algorithm can be considered converged when the change in Q-values between successive iterations becomes very small.

$$\max |Q_{\text{new}} - Q_{\text{old}}| < \epsilon$$

where ϵ is a small threshold such as 0.01.

When the Q-values become stable and further episodes produce negligible changes, the learned policy can be considered converged.

4(c) Final Learned Policy and Ethical Considerations

After Q-Learning converges, the final policy is obtained by selecting the action with the highest Q-value for each state.

Final Learned Policy:

Patient State	Best Action
Low Risk	Monitor
Moderate Risk	Diet / Exercise
High Risk	Medication / Monitor
Controlled	Monitor

The exact action should be selected from the highest learned Q-value in the final Q-table.

Ethical Considerations:

1. **Patient Safety:** The RL agent should not make unsafe treatment decisions. Medical professionals should supervise important treatment recommendations.

2. **Bias:** The training data should represent different patient groups fairly. Bias in the training data can result in unfair or inaccurate recommendations.
3. **Explainability:** The recommendations should be understandable to doctors and patients. The system should provide reasons for the recommended action whenever possible.
4. **Human Oversight:** The AI system should support healthcare professionals rather than completely replacing clinical judgement.

Therefore, patient safety, fairness, explainability and human supervision are essential when deploying reinforcement learning for medical treatment recommendations.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation